to occur remotely. As a result, drones become lighter, more energy-efficient, and capable of more advanced decision-making. Ultimately, high-speed communication serves as the digital nervous system of future drone ecosystems, enabling UAVs to operate not as isolated aircraft, but as coordinated, intelligent, and seamlessly networked components of broader mobility and data infrastructure.

Table 9.2 compares major communication technologies relevant to drone connectivity and coordinated aerial operations. While 4G networks support only basic remote control and short-range missions, 5G introduces high-speed and low-latency connectivity, enabling real-time video streaming, BVLOS operations, and coordinated drone swarms. 6G networks, currently under research, are expected to support ultra-intelligent drone ecosystems capable of autonomous decision-making and citywide aerial mobility integration. Meanwhile, satellite and LEO (Low Earth Orbit) networks expand operational range to remote, maritime, and mountainous environments. Together, these communication systems form the backbone of future adaptive, scalable, and globally connected UAV infrastructures.